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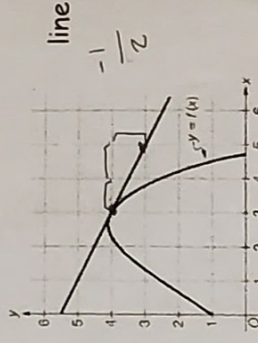
Multiple Choice Questions 1-4 (4)

1. Let f be the function defined by $f(x) = 2x^3 - x$. Which of the following is the average rate of change of f on the interval $[1, 3]$?

- a. $\frac{f(3)+f(1)}{2}$ b. $\frac{f(3)+f(1)}{4}$ c. $\frac{f(3)-f(1)}{2}$ d. $f(3) - f(1)$

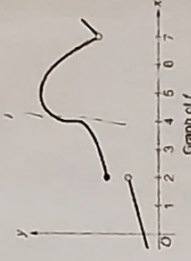
2. Below is shown the graph of the differentiable function f along with the tangent to the graph at $x=3$. What is the value of $f'(3)$?

- a. $-\frac{1}{2}$ b. -2 c. 4 d. $\frac{11}{2}$



3. The figure below shows the graph of a function f , which has a vertical tangent at $x=4$ and a horizontal tangent at $x=5$. Which of the following statements is false?

- a. f is not differentiable at $x=2$ because the graph of f has a jump discontinuity at $x=2$.
 b. f is not differentiable at $x=4$ because the graph of f has a vertical tangent at $x=4$.
 c. f is not differentiable at $x=7$ because the graph has a removable discontinuity at $x=7$.
 d. f is not differentiable at $x=5$ because the graph of f has a horizontal tangent at $x=5$.



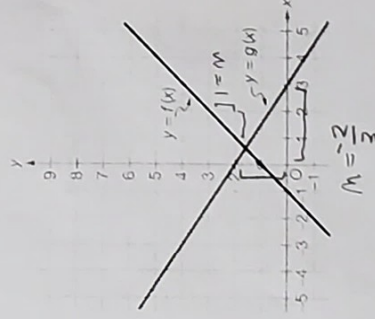
4. Using the table on the right, determine what is the value of $h'(2)$ if $h(x) = \frac{4f(x)}{g(x)+1}$.

- a. -10 b. -7 c. 1 d. 5

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
2	-3	$\frac{1}{2}$	-3	2

Show work for full marks

1. The graphs of the linear functions f and g are shown below. If $h(x) = f(x) + g(x)$, determine $h'(x)$. (2)



$$f(x) = (x-0) + 1$$

$$f(x) = x + 1$$

$$g(x) = -\frac{2}{3}(x) + 2$$

$$h'(x) = \frac{d}{dx}(f(x) + g(x)) = h'(x) = \frac{1}{3}$$

$$h'(x) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

$$h'(x) = 1 + (-\frac{2}{3})$$

$$= h'(x) = \frac{1}{3}$$

2. Differentiate the following using rules for differentiation. Determine $\frac{d^2y}{dx^2}$ in simplified form.

3/4 (4) Expand first then do power rule $y = (x^5 + 1)(3x^2 - x)$ $\rightarrow$ solution?

$$y = 3x^7 - x^6 + 3x^2 - x$$

$$y' = 21x^6 - 6x^5 + 6x - 1$$

$$y'' = 126x^5 - 30x^4 + 6$$

$$\frac{d^2y}{dx^2} = 126x^5 - 30x^4 + 6$$